



An interpretable neural network approach to cause-of-death mortality forecasting

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(Received 27 March 2024; revised 22 August 2024; accepted 28 November 2024)

Abstract

Cause-of-death mortality forecasting, a key topic in public health and actuarial science, is a challenging task due to the difficulty of modeling that accounts for dependencies among causes of death. While several cause-of-death mortality models have been proposed to address this difficulty, little attention has been paid to improving their predictive performance. Recently, purely data-driven approaches using tensor decomposition methods have been introduced to cause-of-death mortality modeling, demonstrating strong out-of-sample predictive performance compared to existing models. However, these methods have difficulties in the interpretability of multi-rank tensor components to achieve strong predictive performance. In response, we propose a novel tensor-based cause-of-death mortality model by replacing the tensor decomposition with a convolutional autoencoder with a one-dimensional latent layer that provides a Lee-Carter-like time-series factor; the model also provides the age sensitivity of cause-specific log mortality to the time-series factor. Due to the representational capability of the neural network, our model achieves better predictive performance compared to the existing tensor decomposition-based models, despite the simplified latent layer for model interpretability.

Keywords: Adaptive penalized tensor decomposition; cause-of-death mortality; convolutional autoencoder

1. Introduction

Cause-of-death mortality forecasting is a key topic in public health and actuarial practice; however, it is a challenging task due to the difficulty of modeling that accounts for dependencies among the causes of death with different evolving patterns. While early studies on cause-of-death mortality modeling assumed independence among causes of death, such as fitting the Lee-Carter (LC) model (Lee & Carter, 1992) to each cause of death, cause-of-death mortality models with interdependencies have been proposed in the current decade using various techniques, including the vector error correction models or cointegration techniques (Arnold & Sherris, 2013, 2015; Arnold & Glushko, 2021), multinomial logistic regression (MLR) method (Alai et al., 2015), forecast reconciliation techniques to ensure coherence between aggregate and cause-specific mortality projections (Li & Lu, 2019), hierarchical Archimedean copula approach (Li et al., 2019), and stochastic age-period-cohort models (Redondo-Loures and Cairns, 2021). Recently, novel data-driven approaches to cause-of-death mortality modeling have been proposed by Huynh & Ludkowski (2024) and Zhang et al. (2023), and our study is in the same context. Huynh & Ludkowski (2024) introduce Gaussian process regression, a Bayesian kernel-based data-driven regression framework that can provide confidence intervals but does not focus on point estimate accuracy; indeed, they do not compare predictive performance with existing models. As a natural

data-driven approach from the perspective of a tensor (i.e., multidimensional array) data format, Zhang et al. (2023) introduce tensor decomposition techniques to cause-of-death mortality modeling, with a motivation based on the acknowledgment that “existing cause-of-death mortality approaches have primarily focused on interpretation and inference, rather than necessarily aiming to improve out-of-sample performance,” with which we agree. Zhang et al. (2023) propose the adaptive tensor decomposition (ADAPT) method, an improvement of the canonical polyadic decomposition (CPD) method, and jointly model the three dimensions (age, year, and cause of death) in a multi-rank tensor form using the method. The ADAPT method reduces the computational burden of the penalized CPD method proposed by Madrid-Padilla & Scot (2017). The out-of-sample prediction of the ADAPT model involves a two-stage estimation applying time-series models to the year factors. On the United States male cause-of-death mortality data, Zhang et al. (2023) demonstrate the strong out-of-sample predictive performance of their ADAPT model compared to MLR, unpenalized CPD, and some existing models, including the cause-specific LC model and the product-ratio model (Hyndman et al., 2013). The product-ratio model accounts for dependencies among causes of death, but the underlying coherence assumption does not allow for different evolving patterns for each cause of death in the mid- and long-term predictions. While the ADAPT model demonstrates strong predictive performance, the multi-rank tensor decomposition defined by cross-validation loses model interpretability. Only the first factor (age, year, and cause of death) can be given some interpretation; however, it is less desirable for a cause-of-death mortality model because the age factor remains the same regardless of the cause of death.

To address the interpretability issue of the multi-rank tensor decomposition, which is required to improve the predictive performance, we propose a novel tensor-based cause-of-death mortality model by replacing the tensor decomposition with a neural network (NN) with a one-dimensional latent layer, aiming to achieve cause-specific LC-like interpretability (i.e., representing age sensitivities of cause-specific log mortality with respect to a single descending time-series factor) without losing the predictive performance. Our model uses a convolutional autoencoder (CAE) architecture, which is widely used for image anomaly detection; it includes convolutional neural networks (CNN) (LeCun et al., 1990), suitable for tensor data processing, and an autoencoder architecture for the dimensionality reduction as an alternative to tensor decomposition. Similar to the ADAPT model, the CAE’s out-of-sample prediction involves a two-stage estimation that applies time-series models to the latent variables in the bottleneck layer of the autoencoder,

In comparison with the ADAPT model, we demonstrate the predictive performance and model interpretability of the CAE model on the World Health Organization (WHO) cause-of-death data. From the data, we select five populations (Japanese males and females, British males and females, and German males) and summarize each population’s data into eight causes. These populations are specifically selected to observe the most recent data (up to 2019) and ensure no zero data in the eight causes of death. The eight selected causes of death include increasing mortality to observe the representability of models for the causes with increasing mortality rate.

The remainder of this paper is organized as follows. Section 2 reviews tensor decomposition techniques for cause-of-death mortality modeling and provides an overview of CAE basics; Section 3 explains the proposed CAE model; Section 4 presents the numerical results of the CAE model in comparison with the ADAPT model. Finally, Section 5 concludes the paper.

2. Preliminaries

2.1 Tensor decomposition for cause-of-death mortality modeling

Zhang et al. (2023) introduce a regularized tensor decomposition technique to cause-of-death mortality modeling to suppress overlearning and improve prediction performance. Regularized tensor decomposition is based on CPD penalized by generalized lasso penalties. An ADAPT method is proposed to overcome the computational challenge of the penalized CPD. Madrid-Padilla & Scot (2017) provide the theoretical backgrounds on the tensors and the CPD.

Let the cause-of-death log mortality data be a tensor denoted by $\log \mathbf{M} \in \mathbb{R}^{n_c \times n_a \times T}$, where n_c, n_a, T are the number of segments of cause-of-death, age, and observation year, respectively. A rank-R CPD aims to decompose the cause-of-death log mortality data into the sum of R-rank-1 tensors (i.e., R outer products of vectors) by solving the problem:

$$\underset{\mathbf{u}_r \in \mathbb{R}^{n_c}, \mathbf{v}_r \in \mathbb{R}^{n_a}, \mathbf{w}_r \in \mathbb{R}^T}{\text{minimize}} \left\| \log \mathbf{M} - \sum_{r=1}^R d_r \cdot \mathbf{u}_r \circ \mathbf{v}_r \circ \mathbf{w}_r \right\|_T, \quad (1)$$

where $0 < d_r \in \mathbb{R}, r = 1, \dots, R$; the vectors $\mathbf{u}_r, \mathbf{v}_r, \mathbf{w}_r, r = 1, \dots, R$ are normalized to unit length for identifiability and correspond to the estimated factor of cause, age, and observation year, respectively; the operation $\circ$ denotes the outer product and $\|\cdot\|_T$ denotes the tensor norm (a dimensional generalization of the Frobenius norm), throughout the paper. The parameters $\mathbf{v}_r, \mathbf{w}_r$ ($r = 1, \dots, R$) have some similarity to the age and time components of the generalized LC model of Renshaw & Haberman (2003) when the model is fitted separately to each cause of death; however, the CPD is applied to all three dimensions of the tensors simultaneously.

Madrid-Padilla & Scot (2017) propose a penalized CPD to promote smoothness and sparsity in tensor decomposition. A rank-R penalized CPD with generalized lasso penalties on the vectors $\mathbf{u}_r, \mathbf{v}_r, \mathbf{w}_r$ is given by solving the problem:

$$\underset{\mathbf{u}_r \in \mathbb{R}^{n_c}, \mathbf{v}_r \in \mathbb{R}^{n_a}, \mathbf{w}_r \in \mathbb{R}^T}{\text{minimize}} \left\| \log \mathbf{M} - \sum_{r=1}^R d_r \cdot \mathbf{u}_r \circ \mathbf{v}_r \circ \mathbf{w}_r \right\|_T + \sum_{r=1}^R (\lambda_{u,r} \|\mathbf{D}_r^u \mathbf{u}_r\|_1 + \lambda_{v,r} \|\mathbf{D}_r^v \mathbf{v}_r\|_1 + \lambda_{w,r} \|\mathbf{D}_r^w \mathbf{w}_r\|_1), \quad (2)$$

where $\lambda_{u,r}, \lambda_{v,r}, \lambda_{w,r}, r = 1, \dots, R$ are the tuning parameters and $\mathbf{D}_r^u, \mathbf{D}_r^v, \mathbf{D}_r^w, r = 1, \dots, R$ are the corresponding penalty matrices. Here, $\|\cdot\|_1$ denotes the L1 norm and the unit norm constraints for $\mathbf{u}_r, \mathbf{v}_r, \mathbf{w}_r, r = 1, \dots, R$ are relaxed to convex constraints (i.e., the norms are less than or equal to one) for optimization without affecting the performance of the decomposition. The larger the R , the better the model fit, but the lower the generalization performance for prediction, so it must be determined by grid search (see Section 4.2). However, selecting the tuning parameter values for $R \geq 3$ is computationally infeasible because the number of combinations of possible sets of tuning parameter values is too large; this problem motivates the introduction of the ADAPT technique proposed by Zhang et al. (2023).

Replacing the penalty matrices $\mathbf{D}_r^u, \mathbf{D}_r^v, \mathbf{D}_r^w$ and tuning parameters $\lambda_{u,r}, \lambda_{v,r}, \lambda_{w,r}$ in Equation (2) with prespecified adaptively weighted penalty matrices $\tilde{\mathbf{D}}_r^u, \tilde{\mathbf{D}}_r^v, \tilde{\mathbf{D}}_r^w$ and three tuning parameters $\lambda_u, \lambda_v, \lambda_w$, the rank-R ADAPT is formulated as

$$\underset{\mathbf{u}_r \in \mathbb{R}^{n_c}, \mathbf{v}_r \in \mathbb{R}^{n_a}, \mathbf{w}_r \in \mathbb{R}^T}{\text{minimize}} \left\| \log \mathbf{M} - \sum_{r=1}^R d_r \cdot \mathbf{u}_r \circ \mathbf{v}_r \circ \mathbf{w}_r \right\|_T + \sum_{r=1}^R (\lambda_u \|\tilde{\mathbf{D}}_r^u \mathbf{u}_r\|_1 + \lambda_v \|\tilde{\mathbf{D}}_r^v \mathbf{v}_r\|_1 + \lambda_w \|\tilde{\mathbf{D}}_r^w \mathbf{w}_r\|_1). \quad (3)$$

In Equation (3), the adaptively weighted penalty matrix $\tilde{\mathbf{D}}_r^u$ and the tuning parameter λ_u are set to zero, since no penalty is needed to promote smoothness and sparsity on the cause-of-death dimension; the other adaptively weighted penalty matrices $\tilde{\mathbf{D}}_r^v, \tilde{\mathbf{D}}_r^w$ are determined using the vector estimates $\tilde{\mathbf{v}}_r, \tilde{\mathbf{w}}_r$ from the unpenalized CPD, as follows.

$$\tilde{\tau}_r^v = \mathbf{D}^v \tilde{\mathbf{v}}_r, \tilde{\tau}_r^w = \mathbf{D}^w \tilde{\mathbf{w}}_r; \tilde{\mathbf{D}}_r^v = \frac{\mathbf{D}^v}{|\tilde{\tau}_r^v|}, \tilde{\mathbf{D}}_r^w = \frac{\mathbf{D}^w}{|\tilde{\tau}_r^w|}, \quad r = 1, \dots, R, \quad (4)$$

where the penalized matrices $\mathbf{D}^v, \mathbf{D}^w$ are

$$\mathbf{D}^v = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix} \in \mathbb{R}^{(n_a-1) \times n_a}, \mathbf{D}^w = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix} \in \mathbb{R}^{(T-1) \times T}. \quad (5)$$

Finally, the tuning parameters λ_v, λ_w are determined by grid search; Zhang et al. (2023) apply five-fold rolling origin cross-validation (Liu & Yang, 2022) to find them.

2.2 Convolutional autoencoder

To replace the tensor decomposition with a NN, we focus on CAE, which combines an autoencoder for dimensionality reduction and a CNN suitable for tensor data processing.

A convolution operation involves an input tensor $z \in \mathbb{R}^{q^{(1)} \times \cdots \times q^{(K)}}$ of order K , where the first $K-1$ components have a spatial structure and the K -th component stands for the channels. The number of channels is a hyperparameter of CNN and is not tied to specific elements of the data, unlike the other components.

For $K=3$, the convolution is a mapping:

$$z^{(1)} = \left(z_{i_1, i_2, j}^{(1)} \right) \in \mathbb{R}^{q_1^{(1)} \times q_1^{(2)} \times q_1^{(3)}} \rightarrow z^{(2)} = \left(z_{i_1, i_2, j}^{(2)} \right) \in \mathbb{R}^{q_2^{(1)} \times q_2^{(2)} \times q_2^{(3)}}. \quad (6)$$

For example, in the later section, the first and second components of the tensors are tied to the age and cause elements of the cause-of-death mortality data, respectively. Using intercepts $b_j \in \mathbb{R}$ and filter weights $W_j = (w_{l_1, l_2, l_3, j}) \in \mathbb{R}^{f^{(1)} \times f^{(2)} \times f^{(3)}}$ for $j = 1, \dots, q_2^{(3)}$, where $f^{(1)} = q_1^{(1)} - q_2^{(1)} + 1$, $f^{(2)} = q_1^{(2)} - q_2^{(2)} + 1$, and $f^{(3)} = q_1^{(3)}$, the exact formula for the mapping is

$$z_{i_1, i_2, j}^{(2)} = \varnothing(b_j + \sum_{l_1=1}^{f^{(1)}} \sum_{l_2=1}^{f^{(2)}} \sum_{l_3=1}^{f^{(3)}} w_{l_1, l_2, l_3, j} z_{i_1+l_1-1, i_2+l_2-1, l_3}^{(1)}), \quad j = 1, \dots, q_2^{(3)}, \quad (7)$$

where $\varnothing$ represents an activation function and the stride (i.e., the distance between two consecutive positions of the filter weight matrices over the input data) is set to one. For simplicity, only the unit stride is used throughout this paper. Convolution operations are typically implemented with a zero-padding operation before and a pooling operation after. Zero padding is used to reshape input tensors without adding any additional parameters, as the convolution operation reduces the size of the output tensors. Pooling is also a mapping that can be expressed by Formula (5), but it uses a $p^{(1)} \times p^{(2)}$ window ($p^{(1)}, p^{(2)} \in \mathbb{N}$) that provides partitions of the spatial part of the tensor and applies a pooling operator to the partitioned data; the most common pooling operators are the following.

$$\begin{aligned} \text{Average - pooling: } z_{i_1, i_2, j}^{(2)} &= \frac{1}{p^{(1)} p^{(2)}} \sum_{l_1=1}^{p^{(1)}} \sum_{l_2=1}^{p^{(2)}} z_{p^{(1)}(i_1-1)+l_1, p^{(2)}(i_2-1)+l_2, j}^{(1)} \\ \text{Max-pooling: } z_{i_1, i_2, j}^{(2)} &= \max_{1 \leq l_1 \leq p^{(1)}, 1 \leq l_2 \leq p^{(2)}} \left\{ z_{p^{(1)}(i_1-1)+l_1, p^{(2)}(i_2-1)+l_2, j}^{(1)} \right\}. \end{aligned} \quad (8)$$

While pooling layers are used with convolution layers for dimensionality reduction, autoencoder neural networks are used more generally for dimensionality reduction. A typical autoencoder network has a bottleneck architecture where the input and output layers have the same dimension (number of neurons), and a dimensionally reduced latent layer (bottleneck layer) is placed between them. The network from the input layer to the latent layer is called the encoder, and the network from the latent layer to the output layer is called the decoder. In an autoencoder network, the network parameters are determined to minimize the reconstruction error of the output relative to the input.

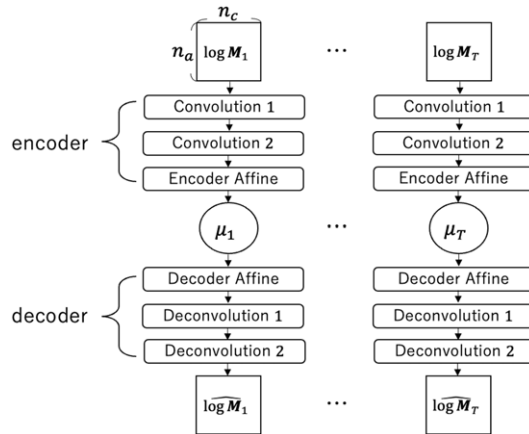


Figure 1. Proposed CAE architecture.

CAE is a neural network that combines the architectural elements of both CNN and autoencoder and is suitable for dimensionality reduction of tensor data. The encoder repeats convolution and pooling, then connects to affine layers (i.e., fully connected layers) to compress the dimensions. The decoder reconstructs the input data using a network where convolution is replaced by deconvolution (transposed convolution), and pooling is replaced by upsampling interpolation (unpooling). Deconvolution and upsampling interpolation increase the dimensionality of the data, whereas convolution and pooling reduce it. For each channel, deconvolution with $f^{(1)} \times f^{(2)}$ weight over $q_1^{(1)} \times q_1^{(2)}$ input data is equivalent to convolution with $f^{(1)} \times f^{(2)}$ weight over $(f^{(1)} + q_2^{(1)} - 1) \times (f^{(2)} + q_2^{(2)} - 1)$ data given by adding zeros over the $q_1^{(1)} \times q_1^{(2)}$ input data, where the output dimension $q_2^{(1)} \times q_2^{(2)}$ is greater than or equal to $q_1^{(1)} \times q_1^{(2)}$ for each axis. Upsampling interpolation is defined so that the mapping of its output with the corresponding pooling function is equal to its input. The nearest neighbor or linear interpolation is commonly used; for example, the nearest neighbor interpolation extends each element of the input into the matrix with the same element that has the same dimension as the corresponding pooling window. For more details on deconvolution, refer to Dumoulin & Visin (2016).

3. Proposed model

To suppress overlearning and improve prediction accuracy, we use a CAE architecture that incorporates an autoencoder architecture for dimensionality reduction and CNNs for learning dependencies among causes of death, where CNNs help reduce the number of parameters through a sparse network structure and convolutional filters shared by all data points.

The cause-of-death log mortality rates for each observation year $t (t = 1, \dots, T)$ form an $n_a \times n_c$ dimensional matrix representing n_a age segments and n_c cause-of-death segments. The CAE model encodes cause-of-death mortality data for each observation year t into a one-dimensional latent variable μ_t and decodes this latent variable into the reproduction data, as shown in Fig. 1. The simplest CAE configuration is adopted as long as the data can be reconstructed, taking into account the computational load. The encoder consists of two convolution blocks and an affine layer, and the decoder consists of an affine layer and two deconvolution blocks. Each convolution block involves zero padding, which adds rows of zeros above and below the input data, convolution with a multichannel filter of size 3×1 , and average pooling with a window of size 2×1 . By setting the dimension for the cause-of-death element to 1, the convolution filter and the

Table 1. Processes and their output dimensions for each observation year t ($t = 1, \dots, T$)

Layer	Process (from top to bottom)	Output dimension
Input		$n_a \times n_c \times 1$
1st Convolution Block	zero padding	$(n_a + 2) \times n_c \times 1$
	convolution using 4 channels of 3×1 window with intercept	$n_a \times n_c \times 4$
	tanh activation	
2nd Convolution Block	average pooling with 2×1 window	$\lfloor \frac{n_a}{2} \rfloor \times n_c \times 4$
	zero padding	$\lfloor \frac{n_a}{2} + 2 \rfloor \times n_c \times 4$
	convolution using 8 channels of 3×1 window with intercept	$\lfloor \frac{n_a}{2} \rfloor \times n_c \times 8$
3rd Convolution Block	tanh activation	
	average pooling with 2×1 window	$\lfloor \frac{n_a}{4} \rfloor \times n_c \times 8$
	tanh-activated affine transformation	1
Decoder Affine	tanh-activated affine transformation	$8n_c \lfloor \frac{n_a}{4} \rfloor$
1st Deconvolution Block	convolution using 4 channels of 3×1 window with intercept	$\lfloor \frac{n_a}{2} \rfloor \times n_c \times 8$
	tanh activation	
	zero padding	$\lfloor \frac{n_a}{2} + 2 \rfloor \times n_c \times 8$
2nd Deconvolution Block	deconvolution using 4 channels of 3×1 window with intercept	$\lfloor \frac{n_a}{2} \rfloor \times n_c \times 4$
	tanh activation	
	zero padding	$(n_a + 2) \times n_c \times 4$
3rd Deconvolution Block	deconvolution using 1 channel of 3×1 window with intercept	$n_a \times n_c \times 1$
	tanh activation	
	log-sigmoid activation	$n_a \times n_c \times 1$

($\lfloor \cdot \rfloor$ indicates the floor function).

pooling widow are designed to learn only the age neighborhood effect, not the cause neighborhood effect, since the order of the causes is meaningless; however, the filter is shared among the causes to learn their dependency. Each deconvolution block performs transposed convolution and upsampling, which are the inverse operations of convolution and pooling, respectively. CAE network parameters are shared across all observation years, and are optimized to minimize the error between the cause-of-death log mortality data, $\log \mathbf{M}_t \in \mathbb{R}^{n_a \times n_c}$ and its reconstruction, $\widehat{\log \mathbf{M}_t}$, for all observation years $t = 1, \dots, T$. Applying L2 regularization with intensity λ to the filter weights $\mathbf{W}_{m,j}^{(conv)}$, $\mathbf{W}_{m,j}^{(deconv)}$ and affine layer weights $\mathbf{W}^{(enc)}$, $\mathbf{W}^{(dec)}$, the loss function is given by

$$\begin{aligned}
 LOSS = \sum_{t=1}^T \left| \log \mathbf{M}_t - \widehat{\log \mathbf{M}_t} \right|_F + \lambda \left\{ \sum_{m=1}^2 \left(\sum_{j=1}^{4m} \left| \mathbf{W}_{m,j}^{(conv)} \right|_2 + \sum_{j=1}^{1+3(2-m)} \left| \mathbf{W}_{m,j}^{(deconv)} \right|_2 \right) + \left| \mathbf{W}^{(enc)} \right|_2 \right. \\
 \left. + \left| \mathbf{W}^{(dec)} \right|_2 \right\}, \quad (9)
 \end{aligned}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_2$ denotes the L2 norm. The intensity λ is a hyperparameter to be determined by validation.

Table 1 provides the details of each process in the proposed CAE model, including the output dimensions.

The predictions of the model are obtained by substituting the predicted latent variables μ_t ($t > T$) into the CAE decoder function, where the predicted latent variables are obtained by fitting time-series models to the latent variables $\mu_1, \dots, \mu_T$ derived from the CAE. As a result of the performance comparison, Autoregressive Integrated Moving Average (0, 1, 0) is adopted as the time-series model.

Table 2. Cause-of-death categories

No.	Cause of death
1	Infectious and parasitic diseases
2	Neoplasms
3	Diseases of the nervous system
4	Diseases of the circulatory system
5	Diseases of the respiratory system
6	Congenital malformations, deformities, and chromosomal abnormalities
7	Injury, disease, and external causes
8	Others

Table 3. Validated parameters of ADAPT model

Country	Gender	R	λ_v	λ_w
Japan	male	8	0.01317	0.00868
	female	11	0.01623	0.01462
UK	male	8	0.00868	0.01623
	female	7	0.01623	0.01069
Germany	male	9	0.01462	0.01462

Table 4. MSE (left) and MAE (right) comparisons between CAE (boldface) and ADAPT for Japan (male, female), UK (male, female), and Germany (male), from top to bottom

Country	Gender	MSE		MAE	
		CAE	ADAPT	CAE	ADAPT
Japan	male	0.065292	0.071878	0.175547	0.189706
	female	0.069508	0.085754	0.207027	0.358504
UK	male	0.053807	0.070150	0.198507	0.375135
	female	0.058592	0.087923	0.218596	0.332840
Germany	male	0.075179	0.077532	0.188122	0.295723

The model can provide age sensitivities by cause, $\beta_{x,c}^{(t)}$ (for age x , cause c , year t) using numerical differentiation of the decoder function $f_{dec}^{(CAE)}$ with respect to the latent variable μ_t as follows.

$$\beta_{x,c}^{(t)} = \frac{\partial}{\partial \mu_t} \left(f_{dec}^{(CAE)} (\mu_t) \right)_{x,c} . \tag{10}$$

4. Numerical analysis

4.1 Data used

In this study, we use the cause-of-death mortality data sourced from the WHO mortality database, where the causes of death are classified according to the International Classification of Diseases (ICD). Despite the revision to ICD11 in 2019, this study exclusively relies on data classified under ICD10, the most recent ICD observable in the database. Notably, we do not undertake data translation between different ICDs due to the impracticability of full translation. The age groups are segmented into 19 age groups from age 0 to ages 85 and over, as in the segmentation of Zhang et al. (2023). While Zhang et al. (2023) classify ICD10 causes into five major causes of death and

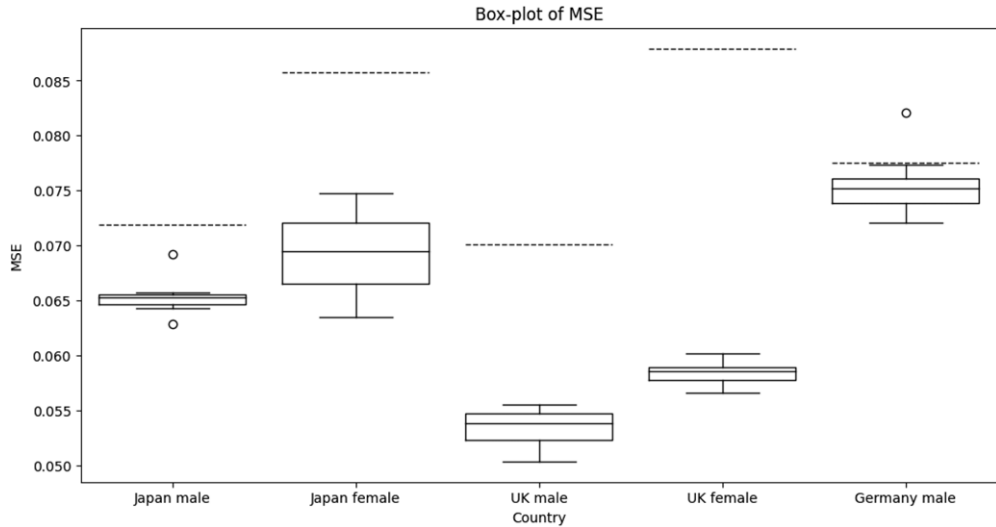


Figure 2. Box plots of CAE’s MSE for Japan (male, female), UK (male, female), and Germany (male), from left to right. Dotted lines indicate ADAPT’s MSE.

others, all categories exhibit a decreasing mortality trend. In this paper, to include categories of causes of death with increasing mortality rates, we initially compress ICD10 categories into the 22 categories of the ICD10-compliant basic classification table of the Japanese Ministry of Health, Labor and Welfare. Subsequently, causes of death categories with zero deaths in any age category are classified as “other” and finally organized into eight categories, as shown in Table 2. As a result of this processing, only five datasets are observable through 2019 without categories of zero deaths: Japan (male and female, 1995–2019), Germany (male, 1997–2019), and the United Kingdom (male and female, 2001–2019). These five datasets are used for the subsequent analyses.

4.2 Model calibration

For each dataset, we use the last five years of data as the test data and the remaining years as the learning data. The ADAPT’s tuning parameters in Equation (3), (λ_v, λ_w) and R , are selected through a grid search using the range of 20 parameters evenly spaced on the log scale from 10^{-6} to 2×10^{-2} for (λ_v, λ_w) , and the range from 3 to 14 for R . While these ranges were the same as in Zhang et al. (2023), our selected parameters did not touch the boundaries of the ranges, as shown in Table 3.

The hyperparameters of the CAE model are determined by validation using the last five years of training data; λ in Equation (9) is set to 0.0005 from the grids {0.05, 0.01, 0.005, 0.001, ..., 0.000001}, and the number of channels for each layer is set as in Table 1. The CAE’s network parameters are optimized using ADAM with an initial learning rate of 0.001 and 15000 learning epochs. PyTorch’s linear interpolation function is used for the upsampling. Network training takes about 40 s per seed using Google Colaboratory, and predictive performance is evaluated as the average of the results for 10 seeds.

4.3 Results

We compare the predictive performance of the CAE model with that of the ADAPT model, which exhibits the strongest out-of-sample predictive performance among the existing cause-of-death

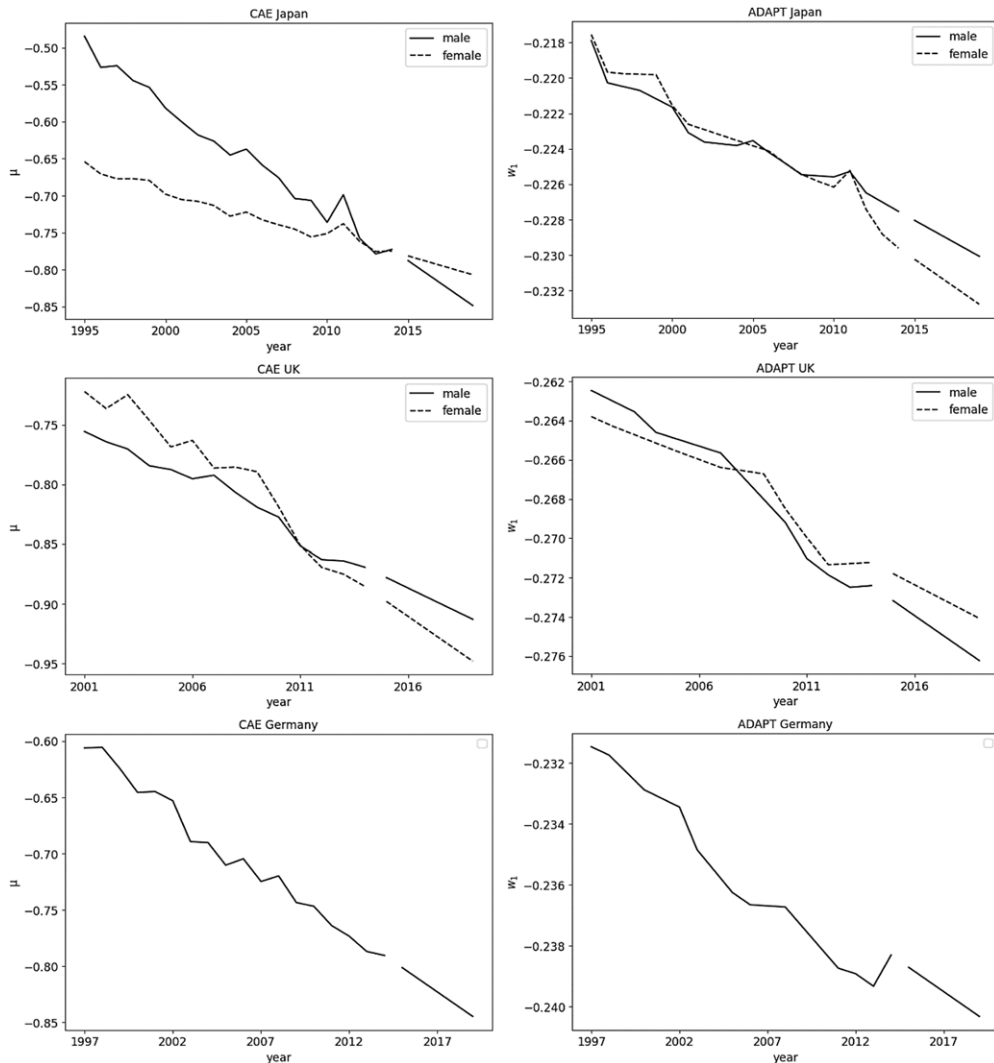


Figure 3. Observed and predicted time-series factors of ADAPT (right) and CAE (left) for Japan, UK, and Germany, from top to bottom. Dotted lines represent females.

mortality models as demonstrated by Zhang et al. (2023). Table 4 shows a comparison of the CAE and ADAPT models in terms of their mean squared errors (MSEs) and mean absolute errors (MAEs) on the test data, with the average values for 10 random seeds for the CAE. The CAE model shows better predictive accuracy than ADAPT for all test data. The seed robustness of the CAE model is illustrated by the box plots for 10 random seeds in Fig. 2, where the dotted lines indicate the MSEs of the ADAPT model, which uses no random seeds.

Next, we compare the interpretation of the CAE and ADAPT models. The ADAPT model has parameter sets $\mathbf{u}_r, \mathbf{v}_r, \mathbf{w}_r, r = 1, \dots, R$ when the rank is R , but we specifically focus on the first factor components $\mathbf{u}_1, \mathbf{v}_1, \mathbf{w}_1$, corresponding to the largest d_r in Equation (3), following Zhang et al. (2023). Figure 3 compares the time-series factor of both models, $\mathbf{w}_1$ for the ADAPT model and the latent variables $\mu_1, \dots, \mu_T$ for the CAE model. Both exhibit LC-like decreasing shapes. While increasing the regularization factor λ_w in Equation (3) simplifies the shape of $\mathbf{w}_1$, the λ_w

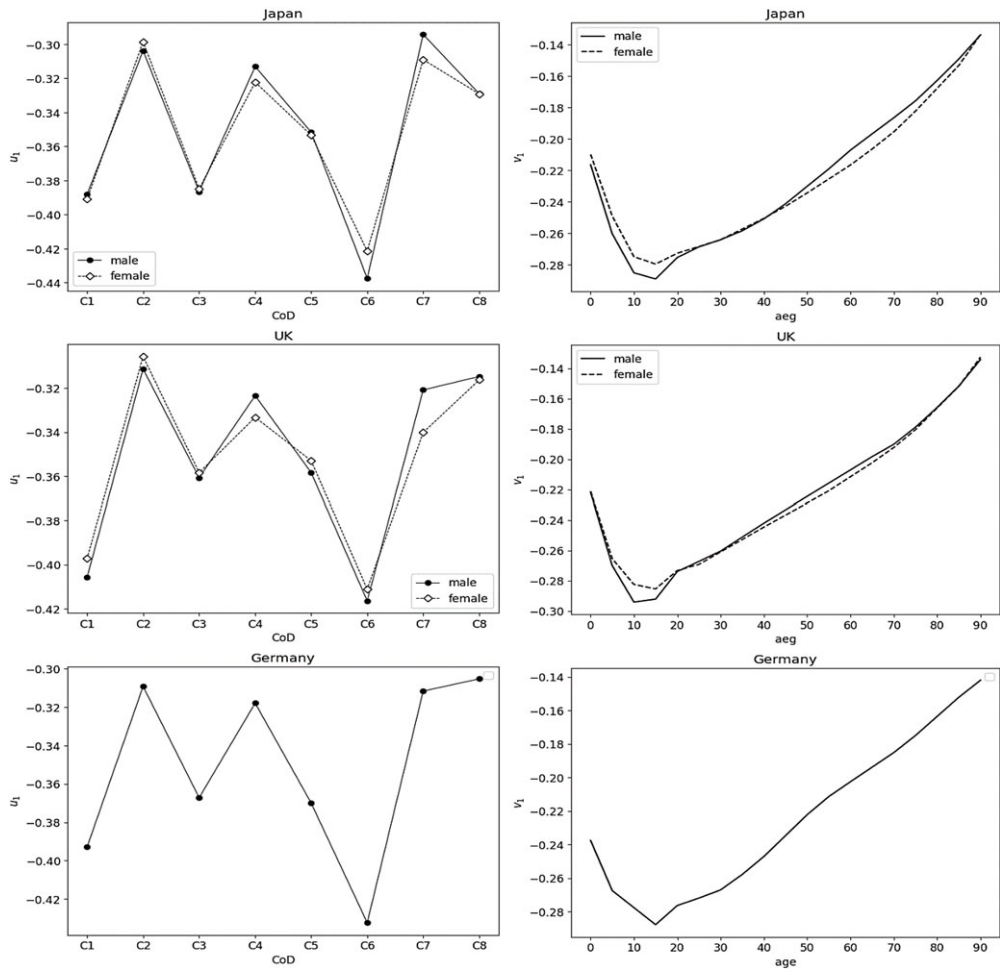


Figure 4. ADAPT's cause-of-death factor u_1 (left) and age factor v_1 (right) for Japan, UK, and Germany, from top to bottom. Dotted lines represent females.

selected by the grid search in Section 4.2 on the data in this paper does not allow w_1 to have simplified shapes as in Zhang et al. (2023). Simplified shapes of w_1 can be obtained by increasing λ_w outside of the grid search range, resulting in larger MSEs than those in Table 4. Examples for UK male and female data are provided in the Appendix A.

Unlike the first time-series factor, the other components of the first factor are not directly comparable to those of the CAE model. In Fig. 4, the first cause-of-death factor u_1 and the first age factor v_1 of the ADAPT model are depicted, where C1 to C8 denote the causes of death corresponding to the numbers in Table 2. Notably, differences by country and gender are small, and the age factors exhibit a mortality-like shape. However, the first factor of the ADAPT model has limitations because the age factors remain the same regardless of the cause of death and year of observation, which may indicate the average age factor across all causes, but does not fit the purpose of cause-of-death mortality models.

Figure 5 shows the second and third factor vectors of the ADAPT model for the Japanese male data, highlighting challenges in model interpretation. While only the first factor vectors of the

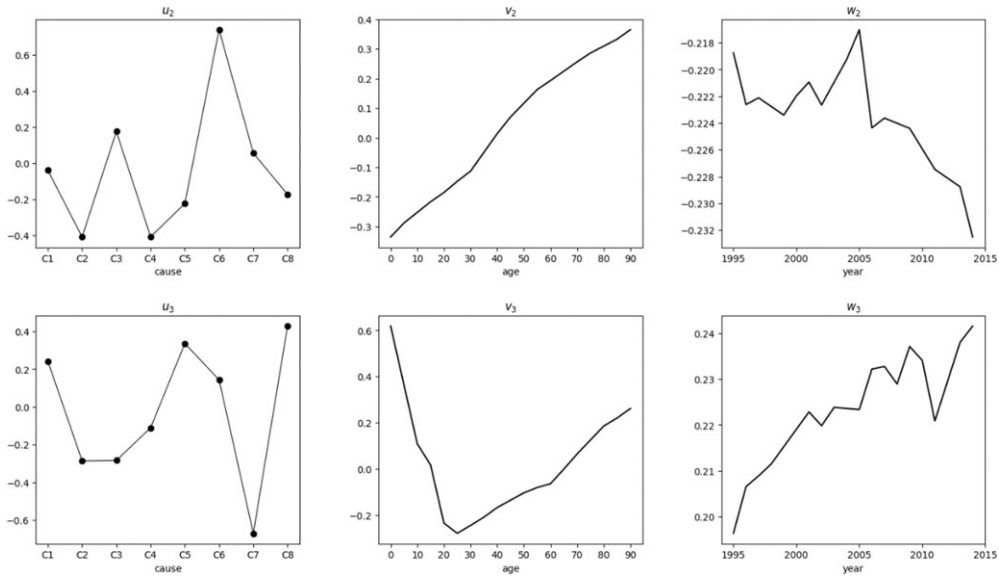


Figure 5. ADAPT's u_2 , v_2 , w_2 (top) and u_3 , v_3 , w_3 (bottom) for Japan male data.

ADAPT model are interpretable, the three vectors take only negative values, as shown in Figs. 3 and 4, and cannot represent causes of death with increasing mortality.

The CAE model has the advantage of the model interpretability by representing $\beta_{x,c,t}$, which is the decoder sensitivity to the time-series factor by cause of death and age. In contrast to the ADAPT model, the CAE model can represent age factors by cause of death and illustrate increases in mortality for specific causes. Figures 6 and 7, for C3 and C4, respectively, show the age sensitivities $\beta_{x,c}^{(t)}$ in Equation (10) and the corresponding observed cause-of-death log mortality data over the training period (up to 2015), with the color of the lines darkening as time approaches 2015.

For C3 (diseases of the nervous system), negative values of the age sensitivities of the CAE for the elderly in each population indicate increasing mortality and are consistent with the observed elderly mortality curves for C3 darkening from bottom to top, as shown in Fig. 6. This is also consistent with the fact that C3 includes Alzheimer's disease, and Alzheimer's disease and other dementias, the second leading cause of death in high-income countries (including all populations in this paper) as of 2019, has experienced the largest increase in deaths since 2000 among the top ten causes of death according to the WHO's Global Health Estimates (GHE) 2000–2019. For C4 (diseases of the circulatory system), positive values of the age sensitivities of the CAE for all ages in each population indicate decreasing mortality, and are consistent with the observed mortality curves darkening from top to bottom, as shown in Fig. 7. This is also consistent with the fact that ischemic heart disease, the leading cause of death in high-income countries as of 2019, has experienced the largest decrease in deaths since 2000 among the top ten causes of death according to the GHE 2000–2019.

Unlike the rank-R ADAPT model, which has R-rank tensor forms and requires R-dimensional time-series models for the projection, the CAE model requires only a single time-series model for the projection due to the nonlinear representability of the network. This gives the CAE model advantages in long-term projections, as shown in Figs. 8 and 9: the 50-year projections by CAE and ADAPT for eight causes of mortality (Fig. 8) and all-cause mortality (Fig. 9) from Japanese male data with dark-colored projection curves and light-colored reconstruction curves. Due to the linearity of the model, the ADAPT model shows changes in projected mortality about twice as large

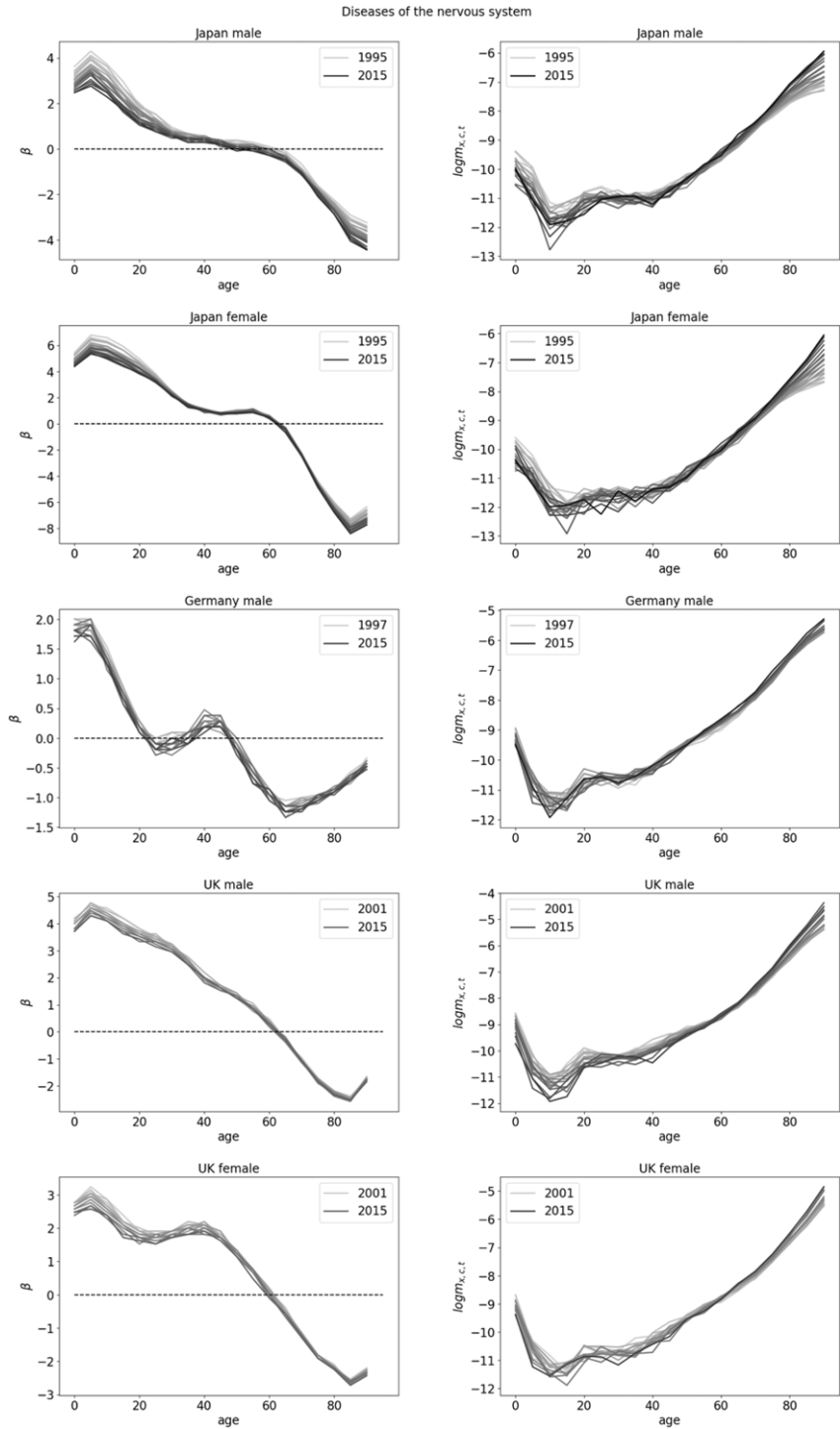


Figure 6. Age sensitivities of CAE (left) and observed mortalities (right) of C3 over the training data (up to 2015) from Japan (male, female), Germany (male), UK (male, female), from top to bottom.

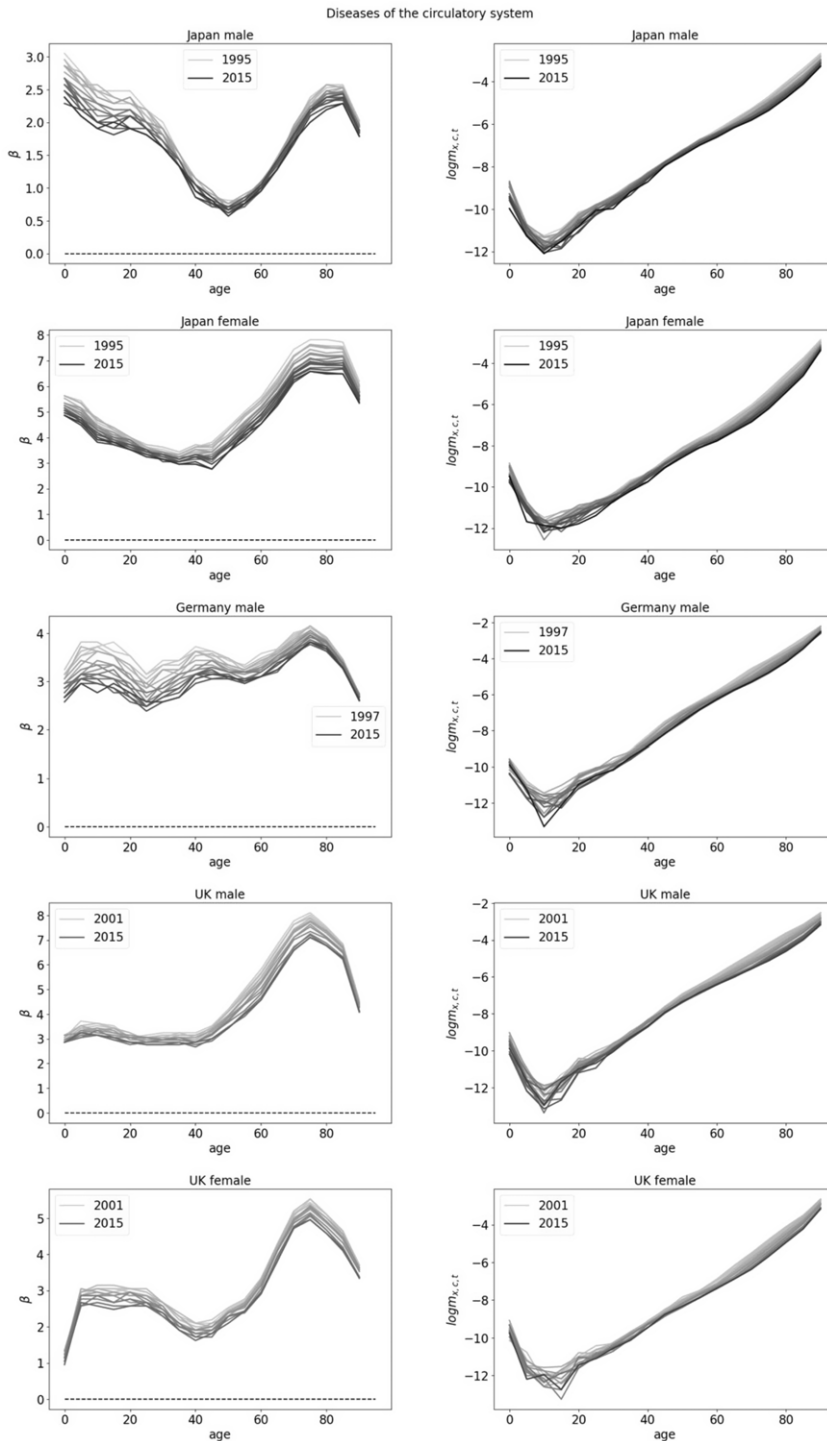


Figure 7. Age sensitivities of CAE (left) and observed mortalities (right) of C4 over the training data (up to 2015) from Japan (male, female), Germany (male), UK (male, female), from top to bottom.

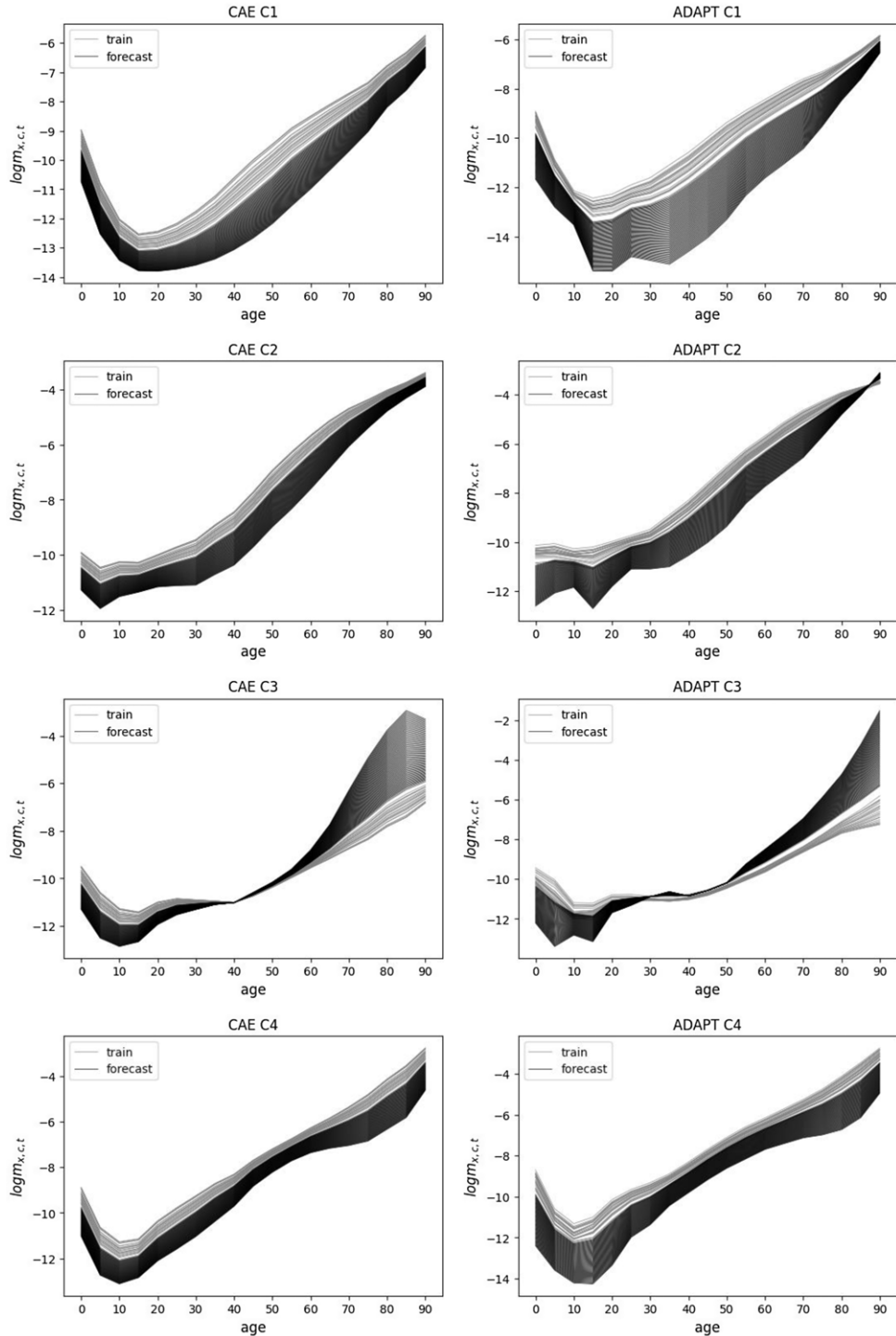


Figure 8. Reconstruction (light lines) and 50-year projection (dark lines) of eight causes of death (C1 to C8 from top to bottom) mortalities using CAE (left) and ADAPT (right) for Japanese male data.

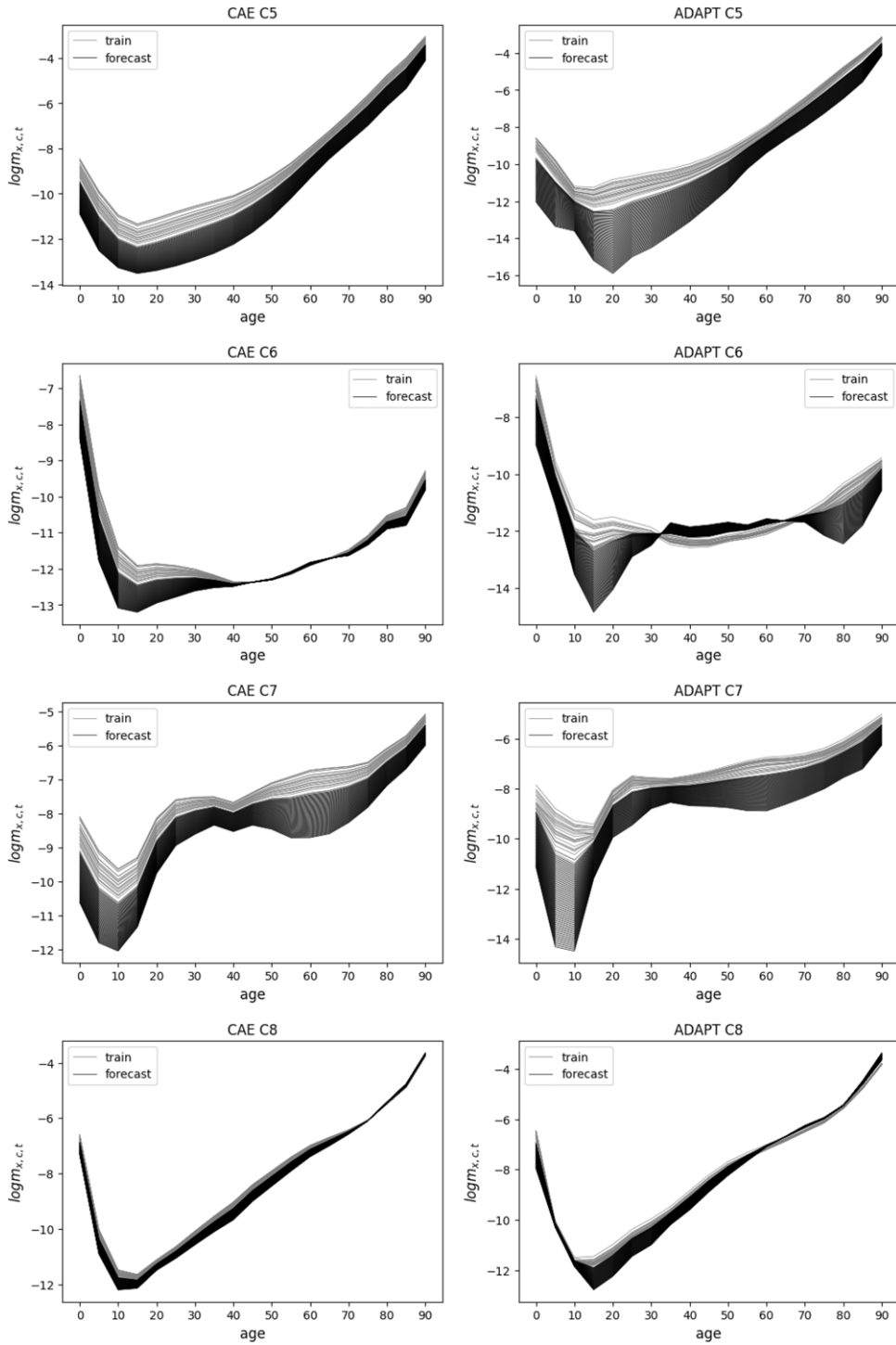


Figure 8. Continued.

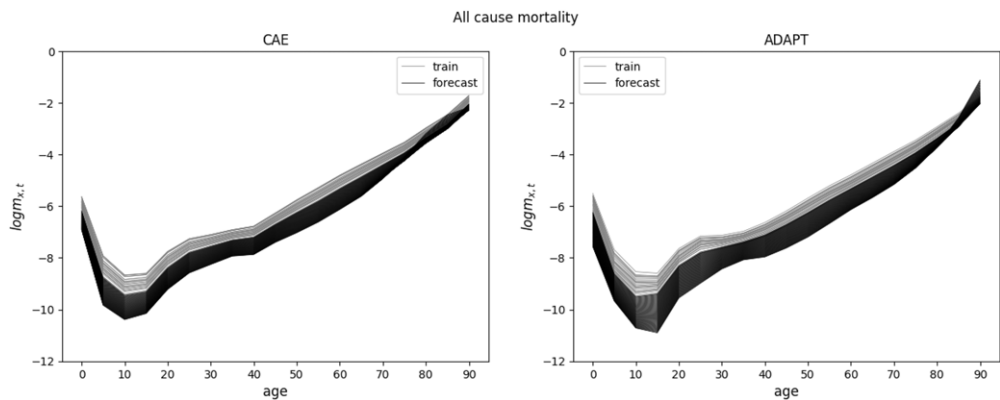


Figure 9. Reproduction (light lines) and 50-year projection (dark lines) of all-cause mortality using CAE (left) and ADAPT (right) for Japanese male data.

as the changes in reconstructed mortality, proportional to the projection period being twice as long as the training period, but the CAE model shows smaller changes in projected mortality due to the nonlinearity of the model and is more consistent with the observed slowdown in mortality improvement (see, e.g., Djeundje et al., 2022). Unlike the CAE model, the projected cause-of-death mortality curves with the ADAPT model lose the smoothness observed in the reconstructed curves that reflects the similarity of the cause-of-death mortality at close ages in the observations. In particular, for causes C1, C2, and C3, the ADAPT model shows unexplainable differences by age in the projected cause-of-death mortality.

5. Conclusion

While the tensor decomposition-based cause-of-death mortality models, including ADAPT, are a natural data-driven approach from a data format perspective, they have an interpretability problem due to the multi-rank tensor decomposition to achieve strong predictive performance. To address this issue in the context of tensor-based cause-of-death mortality modeling, we propose to replace the tensor decomposition with a CAE with a one-dimensional latent layer to achieve LC-like interpretability of the model; despite the simplified latent layer for interpretability, the loss of model representability is limited due to the nonlinearity of the CAE network, resulting in better out-of-sample predictive performance than the ADAPT model on WHO cause-of-death data from five populations. Expanding the dimensionality of the latent layer of the CAE could improve the predictive performance, but at the expense of the interpretability we focus on. Rather, a deeper configuration of the CAE may improve the predictive performance if computational resources are available. The CAE architecture also contributes to more natural results than ADAPT in long-term projections, in the sense that the smoothness and changes in the projected cause-of-death mortality curves by CAE are more consistent with the smoothness in the reconstructed curves and the observed slowdown in mortality improvement than those projected by ADAPT.

This model produces interesting age sensitivity curves that vary by country, gender, and cause of death. However, the analysis of medical implications remains a topic for future research. Similar to the ADAPT model, our model has a limitation of two-stage estimation, and improving the model for single-stage estimation is another topic for future work.

Data availability statement. The data and codes supporting the results of this study are available in the WHO mortality database at <https://www.who.int/data/data-collection-tools/who-mortality-database> and in Appendix B.

Funding statement. This study was supported by the Institute of Science and Technology, Meiji University.

Competing interests. None.

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Appendix A

Figure A.1 shows ADAPT's first factor components $\mathbf{u}_1$, $\mathbf{v}_1$, $\mathbf{w}_1$ corresponding to $\lambda_w = 0.145374$ from the UK male and female data, where their MSEs (0.09179 for male, 0.10003 for female) on the test data are larger than those in Table 4 corresponding to the grid-searched λ_w (0.01623 for male, 0.01069 for female).

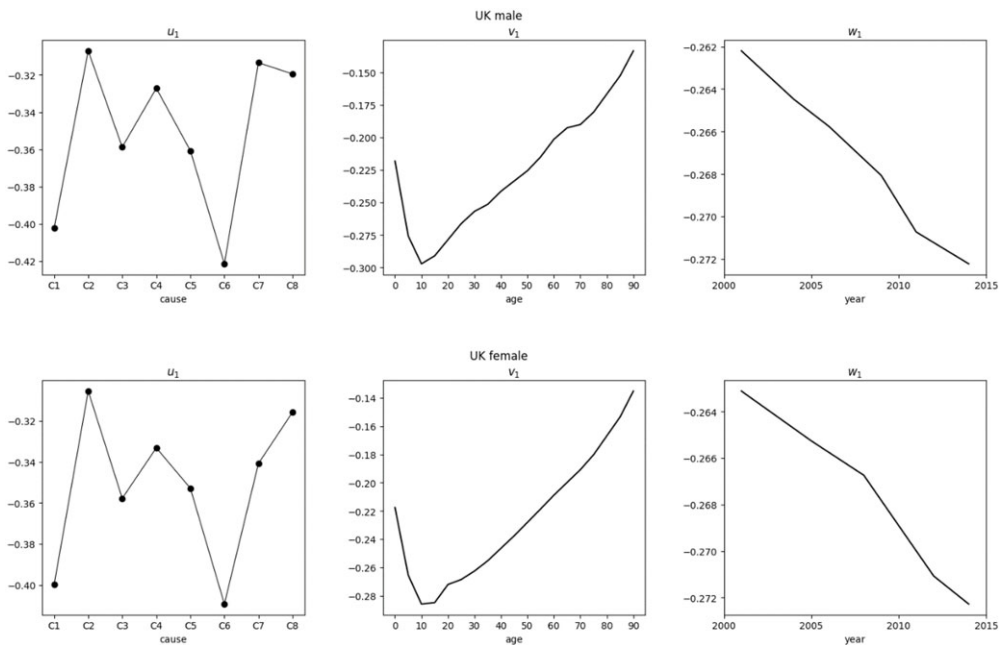


Figure A.1. ADAPT's u_1 , v_1 , w_1 (from left to right) corresponding to $\lambda_w = 0.145374$ from UK male (top) and female (bottom) data.

Appendix B

```

import torch
from torch import nn
from torch.nn import init
import numpy as np

device = torch.device('cuda' if torch.cuda.is available() else 'cpu')

class CAE(torch.nn.Module):
    def __init__(self, num cause=8, num age=19, num kappa=1, num conv=2, num affine=1,
bias=False, out channels=[1, 2, 4, 8]):
        super().__init__()

        #nums
        self.num cause, self.num age, self.num kappa = num cause, num age, num kappa
        self.num conv, self.num affine, self.flatten, self.bias, self.out channels =
num conv, num affine, torch.nn.Flatten(), bias, out channels

        #modules
        self.conv, self.enc affine, self.dec affine, self.deconv = [], [], [], []

        #conv module
        for i in range(self.num conv):
            #conv block
            conv layer = torch.nn.Conv2d(
                in channels=self.out channels[i],
                out channels=self.out channels[i+1],
                kernel size=(1, 3), stride = 1, padding=(0, 1), bias=self.bias
            )
            self.conv.append(conv layer)
            self.conv.append(torch.nn.Tanh())
            self.conv.append(torch.nn.AvgPool2d(kernel size = (1, 2), padding=0))

            #deconv block
            conv tr layer = torch.nn.ConvTranspose2d(
                in channels=self.out channels[i+1],
                out channels=self.out channels[i],
                kernel size=(1, 3), stride = 1, padding=(0, 1), bias=self.bias)
            self.deconv.append(conv tr layer)
            self.deconv.append(torch.nn.Tanh())
            self.deconv.append(torch.nn.Upsample(size = (self.num cause, int(self.num age *
(1/2) ** i)), mode='bilinear'))

        self.num = int(self.num age * (1/2)**self.num conv) * self.num cause *
self.out channels[self.num conv]
        self.deconv.reverse()

        self.conv, self.deconv = nn.ModuleList(self.conv), nn.ModuleList(self.deconv)

        #affine module
        if self.num affine == 1:
            self.enc affine.append(torch.nn.Linear(self.num, 1))
            self.enc affine.append(torch.nn.Tanh())
            self.dec affine.append(torch.nn.Linear(1, self.num))
            self.dec affine.append(torch.nn.Tanh())
        else:
            for i in range(self.num affine-1):
                neurons = int(self.num * (1/2)**i)
                #enc affine
                self.enc affine.append(torch.nn.Linear(neurons, int(neurons/2)))
                self.enc affine.append(torch.nn.Tanh())
                #dec affine
                self.dec affine.append(torch.nn.Tanh())
                self.dec affine.append(torch.nn.Linear(int(neurons/2), neurons))
            self.enc affine.append(torch.nn.Linear(int(neurons/2), 1))
            self.enc affine.append(torch.nn.Tanh())
            self.dec affine.append(torch.nn.Tanh())
            self.dec affine.append(torch.nn.Linear(1, int(neurons/2)))
            self.dec affine.reverse()

        self.enc affine, self.dec affine = nn.ModuleList(self.enc affine),
nn.ModuleList(self.dec affine)

        #mort -> mort reconstruct
        def forward(self, x):
            for layer in self.conv:

```

```

        x = layer(x)
    x = self.flatten(x)
    for layer in self.enc affine:
        x = layer(x)
    for layer in self.dec affine:
        x = layer(x)
    x = x.reshape(x.shape[0], self.out channels[self.num conv], self.num cause,
int(self.num age * (1/2) ** self.num conv))
    for layer in self.deconv:
        x = layer(x)
    return x

#mort -> mu
def enc output(self, x):
    for layer in self.conv:
        x = layer(x)
    x = self.flatten(x)
    for layer in self.enc affine:
        x = layer(x)
    return x

#mu -> mort predict
def dec output(self, x):
    for layer in self.dec affine:
        x = layer(x)
    x = x.reshape(x.shape[0], self.out channels[self.num conv], self.num cause,
int(self.num age * (1/2)**self.num conv))
    for layer in self.deconv:
        x = layer(x)
    return x

```